

Understanding Novice Users' Mental Models of Gesture Discoverability and Designing Effective Onboarding

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ABSTRACT

A variety of consumer Augmented Reality (AR) applications have been released on mobile devices and novel immersive headsets over the last five years, creating a breadth of new AR-enabled experiences. However, these applications, particularly those designed for immersive headsets, require users to employ unfamiliar gestural input and adopt novel interaction paradigms. This leap forward intensifies the complexity of help-seeking and onboarding needs for the end-users. Recent emergence of artificial intelligence (AI)-powered in-context help tools has become potential alternatives to onboarding and search methods. However, non-technical users struggle with prompt-based interactions within LLMs that offer human-like language capabilities, which is unique, but can also be unreliable. My doctoral research aims to (1) understand how novice users discover gestural interactions and classify the types of interaction challenges they face; (2) investigate the nuances in users' mental models of emerging technologies, such as LLMs and AR; and, (3) explore the design of onboarding that enhances gesture discoverability and their application within the AR environments.

CCS CONCEPTS

• Human-centered computing → Empirical studies in HCI.

KEYWORDS

augmented reality; mixed reality; gesture discoverability; help-seeking; large language models; prompt-based interactions

ACM Reference Format:

Anjali Khurana and Parmit K Chilana. 2024. Understanding Novice Users' Mental Models of Gesture Discoverability and Designing Effective Onboarding. In *Companion of the 2024 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp Companion '24)*, October 5–9, 2024, Melbourne, VIC, Australia. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3675094.3678370>

1 PROBLEM STATEMENT

The exploration of Augmented Reality (AR) applications have witnessed significant growth [3] across various domains, but still the widespread adoption of AR applications remains a challenge [9, 41].

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UbiComp Companion '24, October 5–9, 2024, Melbourne, VIC, Australia

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ACM ISBN 979-8-4007-1058-2/24/10

<https://doi.org/10.1145/3675094.3678370>

Unlike conventional interfaces such as WIMP (Windows, Icons, Menus, Pointer) and interactions based on keyboards/mice, AR introduces users to a myriad of novel interaction paradigms, including geolocated egocentric navigation and mid-air gestures, where users can interact with the user interface without touching or holding a physical device [12]. Users must transfer their understanding of desktop and mobile computing to seamlessly interact with interfaces that overlay onto the real world. The complexity intensifies in the domain of immersive AR hosted on head-mounted displays (e.g., Microsoft HoloLens, Magic Leap), where sensors and cameras are used to seamlessly integrate 3D virtual objects into the user's immediate physical surroundings, blurring the boundaries between the digital and physical realms [6, 13, 21]. An unresolved challenge for AR interaction remains; whether a new user can *discover the correct gestural input* required to use an AR system ([4, 21, 32, 48, 59, 61]) given the *conceptual and cognitive interaction* challenges novices may experience [35] that lead to faulty mental models, particularly for novices. This leap forward intensifies the onboarding needs, especially for non-technical users.

When novice users struggle to learn any feature-rich application on their own, they currently turn to various software help resources to learn and seek help for such software tasks. The recent emergence of Generative AI and pre-trained Large Language Model (LLM)-based assistants like *ChatGPT* [2, 16] offers a novel approach to support end-users' software help-seeking by leveraging these assistant's advanced natural language understanding [64]. The increasing interest in LLMs has inspired several initiatives to utilize their capabilities for assisting users with software tasks [1, 56]. For example, some experimental work is being explored by integrating LLMs directly into feature-rich applications, such as Copilot in Microsoft 365 [56] and Firefly in Adobe [1]. However, the effectiveness of LLM-based assistance and how end-users employ LLMs to seek help for feature-rich applications remain important open questions.

Overall, my PhD research focuses on understanding users' mental models when employing novel interaction paradigms, such as gestural-based interactions in AR and prompt-based interactions with LLMs and inventing onboarding and assistance techniques for novice users of AR and LLMs. This understanding is crucial, particularly for novel situations and emerging technologies like AR, which introduce unique interaction paradigms and require users to translate their knowledge from traditional desktop and mobile interfaces, presenting additional conceptual and cognitive challenges that lead to faulty mental models, particularly for novices. The research questions guiding this exploration were:

- **RQ1:** How do novice users discover gestural input and what interaction challenges do they face while interacting with unfamiliar immersive AR applications?

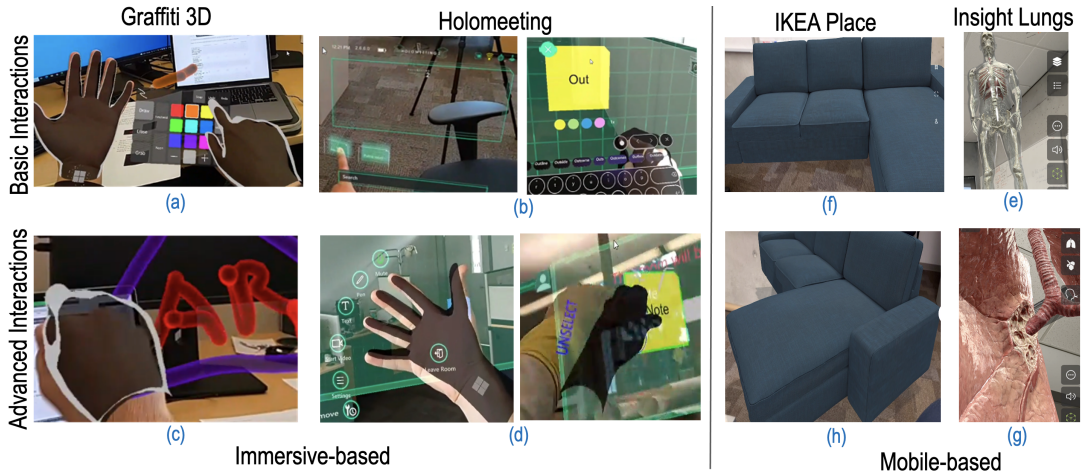


Figure 1: Tasks and Interactions (Participant View): *Basic Interactions:* (a) Selecting the draw option from a menu, (b) Navigating to a room and adding a sticky note, and (e,f) Calibrating virtual objects (e.g., a sofa or the human body); *Advanced Interactions:* (c) Grabbing a sketch, (d) Moving a sticky note, and (g,h) Changing the size and seeing the structure of virtual objects

- **RQ2:** How do end users make use of prompt-based interactions when finding software help and identify gaps in users' mental models as they try to seek and apply LLM-generated software help for their tasks?
- **RQ3:** Based on these insights, how can we design onboarding in AR environments for enhancing the gesture discoverability?

2 RELATED WORK

2.1 Challenges When Designing and Using Gestural Interactions

Research in AR has, so far, acknowledged issues and breakdowns related to technical challenges [5, 11], precision issues [15, 49, 50], system errors [5, 61], social acceptance challenges [5, 11, 36], privacy concerns [26], and fatigue while using gestural interaction [7, 38, 40, 43, 55, 57]. Extensive research has explored efficient interaction techniques [15], and devised alternative ways of interacting in AR (e.g., using a user's body as an input device [36, 42, 52, 63], using external devices [18, 60, 65], etc.) to resolve precision issues [15, 49, 50, 52] and fatigue problems [7, 38, 40, 43, 52, 55, 57]. However, issues related to the learnability of gestures in AR [10, 17, 61] and the cognitive impact of user errors [24, 35] are only starting to be recognized. In fact, Norman and Nielsen [45] argue that gestural interfaces are a step "backward in usability" as fundamental interaction design principles are often not considered. Considering AR applications are a tremendous leap from the desktop and mobile-based interfaces that most users are familiar with [13], understanding user experiences and interaction challenges and designing discoverable interaction techniques is necessary to ensure the long-term adoption and usefulness of these applications [31]. The present work complements the existing research by conducting an in-depth error analysis of gestural interaction and by examining novice users' gulfs of execution and evaluation and other proposed guidelines for gestural interactions [45]. We contribute new knowledge about how novice users' experiences with AR applications

led them to discover gestural interactions and identified when they encountered breakdowns during different task-specific gestures.

2.2 Evolution in Learning and Onboarding Within Complex Applications

HCI research has a rich history of investigating the challenges that users experience when learning and seeking help for complex feature-rich applications across 2D and 3D domains [19, 23, 29, 46]. Help-seeking resources and approaches have evolved over the years: from formal documentation and manuals [47, 51] to the use of videos [30, 34], interactive tutorials [46], Google Search, Q&A or FAQ sites, blogs, dedicated forums [29] and even contextual help systems embedded within applications [14, 19, 22, 25, 33]. However, recent developments in Generative AI have opened a new outlet of help-seeking for users through LLM-based assistance, such as ChatGPT, which is much more forgiving in letting users describe their queries using natural language [53, 64]. These assistants tend to provide more specific and in-context assistance [64] to users, unlike traditional resources that may be scattered and require precise queries for retrieval. Another challenge for end-users, especially novices and non-AI experts, is to define the appropriate prompts and learn prompting strategies [8, 53, 62, 64] to get the desired assistance from these LLMs.

In the specific context of software help-seeking, some experimental work is being explored by integrating LLMs directly into feature-rich applications, such as Copilot in Microsoft 365 [56] and Firefly in Adobe [1]. While there is a rich history of work supporting software learnability and easing help-seeking processes [19, 20, 22, 34], it is unclear whether users' learning approaches and strategies apply to LLM-based assistants and user onboarding in AR, which has distinct interaction paradigms. Considering LLMs are a tremendous leap from traditional help-seeking mediums that most users are familiar with, there have been calls to investigate users' mental models as they interact with LLMs [37]. This becomes necessary when seeking assistance for feature-rich software tasks, where there is an interplay between the mental model of LLM vs

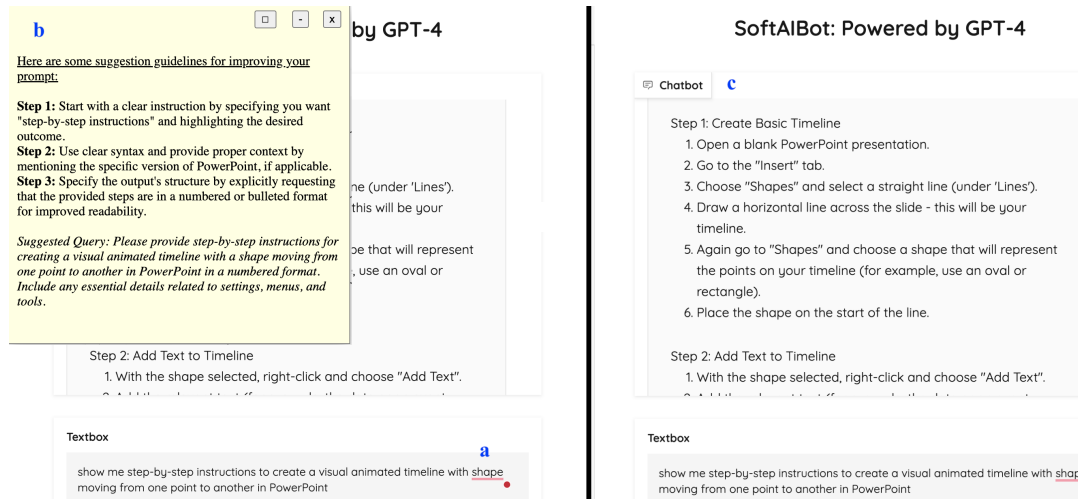


Figure 2: SoftAIBot integrates domain context via documentation and offers prompt guidelines to construct better prompts: (a) allows users to type in the prompt text and submit it; (b) generates prompt suggestions in-response to a user's text (in this case, also shows a sample transformed query that users can directly use); (c) formats the response as step-by-step instructions optimized for particular software contexts, in this case PowerPoint.

the software application [29, 37]. Furthermore, to harness the full potential of these LLM assistants for seeking help for their software tasks, we need more insights on where and how users struggle with these LLMs [39] in seeking accurate and relevant assistance. We extend this prior work by examining how novices employ LLM assistants for software help-seeking and the types of challenges they experience. Our study contributes new knowledge on how non-AI expert end-users employ LLMs' generated software guidance assistance in accomplishing tasks for feature-rich applications by comparing our own implemented, SoftAIBot, an LLM optimized for particular domain context with the Baseline ChatGPT.

3 METHODOLOGY AND EVALUATION

3.1 Understanding how novice users discover appropriate gestural input techniques with consumer AR application

To answer our research questions, in the first part of my research, we ran a two-part study: (1) *observational user study* (2) *expert validation study*. First, we conducted an *observational study* with 25 participants from diverse backgrounds, occupations, and technical knowledge to empirically study the first-time onboarding experience of novice users and establish a better understanding of the interaction challenges they face when interacting with immersive AR applications, specifically head-worn AR. We used a within-subject design wherein each participant used four AR applications, comprising both immersive-based and mobile-based AR hosted on Microsoft Hololens 2 and Google Pixel 6/iPad mini, respectively. Participants started first with interactions using basic gestures involving unimanual gestures (e.g., selection, typing, creating a virtual 3D sketch on a real-world object such as a laptop in Figure 1c) and then used advanced gestures involving bimanual gestures (e.g., access menus, grab and rotate objects). We considered applications

and tasks (see Figure 1) that would require different interaction techniques (e.g., voice, typing, gestures, touch interactions) that could lead to different types of AR breakdowns. We encouraged participants to think aloud and also conducted follow-up interviews to further probe any difficulties that impacted the use of each application. To investigate the mental models that participants constructed and understand how participants discovered the different gestures that were needed to interact with AR, we used Norman's action cycle, fundamental cognitive principles of discoverability and interaction [44] and fundamental Norman-Nielsen (NN) principles of interaction design relevant to gestural interfaces [45]. To analyze how participants initially discovered the correct gestures to use, we analyzed each participant's gesture trial attempts for major input task actions (i.e., Selection, Typing, Getting a menu, Grabbing).

Next, to contextualize the prevalence of our findings and their practical implications, we carried out an *expert validation study* with eight professional AR designers, engineers, and researchers from four large technology companies. Our findings showed that the majority of participants largely struggled to discover the correct gestural inputs and relied on trial-and-error explorations. A detailed analysis of gesture interaction traces in each task revealed that few errors result from system recognition issues and that the majority of errors occurred when participants could not determine the correct sequence of actions to perform (i.e., *gulf of execution*) or could not evaluate their actions (i.e., *gulf of evaluation*). Reflecting on their real-world experiences, these experts confirmed that the discoverability issues revealed in our study are significant hurdles for nearly all newcomers to AR gestural interfaces. While the experts discussed potential improvements in low-level sensing and gesture recognition technologies to address these challenges, they collectively underscored the importance of accommodating users' diverse mental models and addressing user errors during the design

phase. The experts emphasized the importance of designing appropriate onboarding approaches that are natural and integrated into the application to facilitate better learnability of AR user interfaces.

3.2 Investigating how software users make use of LLMs with *prompt-based interactions* for seeking help for feature-rich applications

LLM-based assistants have become a new outlet for user onboarding and help-seeking for diverse software applications. In the second part of my research, I expanded my work to investigate the nuances and users' mental models when employing prompt-based interactions with LLMs in concert with feature-rich software tasks. In particular, we investigate the effectiveness of recently emerging prompt guidelines [54, 58, 64] (offered by OpenAI, Microsoft, etc.) and considered the impact of integrating additional domain context, such as software documentation, into LLMs to improve the precision of the LLM output. For this investigation, we developed *SoftAIBot* (See Figure 2), our implementation of a state-of-the-art LLM assistant that integrates prompt guidelines and domain context (e.g., software documentation), and compared it with a Baseline ChatGPT Plus LLM assistant. We ran a controlled within-subject experiment study and follow-up interviews with 16 non-ML expert participants to assess the overall experience of users seeking assistance from LLM interventions for software tasks (e.g., PowerPoint, Excel) along with their perceptions of accuracy, relevancy, ease of use, and trust of the assistance provided by LLM intervention. To measure the participants' success in applying LLM assistance and how accurately the LLM provided instructions, our metrics also included task completion rate, task accuracy rate, and accuracy and relevancy of LLM Assistance. By analyzing users' interactions, behaviors, and challenges when using these state-of-the-art LLMs for software help-seeking, we aim to help end users harness the full potential of LLM-based assistants for feature-rich software [39] and further for inventing onboarding assistance for AR applications.

We hypothesized that our implementation of *SoftAIBot* would generate accurate and relevant assistance for users' prompts and help users finish their software tasks. However, our findings showed that even though *SoftAIBot* performed better than BaseLine in generating more accurate and relevant LLM output, users could not recognize these differences, leading to poor task completion. Our findings illustrate the challenges that software users experience with prompt-based interaction (e.g., crafting prompts, comprehending how prompts bias LLM output, mapping LLM-suggested steps to software, overtrusting output correctness) and identify gaps in users' mental models as they try to seek and apply assistance from LLMs to software tasks, which affect both their use of LLMs and perception of software features, regardless of implicit enhancements in the underlying model or explicit prompt guidelines (*SoftAIBot*). Lacking an accurate mental model of LLMs, users tended to over-trust the LLM assistance without much contemplation. We discuss the implications of these findings for designing and implementing LLM help tailored to feature-rich applications, the need to incorporate transparent and responsible LLM assistants, and ways to bridge the disparity between mental models and LLM interfaces, ultimately helping end-users form accurate mental models of LLMs.

3.3 Proposed Study and Evaluation

The findings from my prior works have important implications for the design of more intuitive gestural input and interaction, as well as the development of streamlined help that can facilitate AR-specific onboarding. Given the potential of AI in enhancing help-seeking behaviors, I propose a subsequent study that builds upon insights and nuances from both of my previous research. This study aims to design onboarding in AR environments that employs a user-centered approach and guides users to discover and employ complex gestural interactions for their tasks. The study will help us understand (a) whether interactive onboarding instead could lower the barriers to properly discovering and learning AR gestural interaction, (b) what could be a useful way to design the onboarding in AR environments by leveraging the user-led errors, and (c) real-world deployments to assess discoverability and learnability of AR gestural interactions when using the proposed onboarding (d) users' perceptions of usefulness, relevancy, and accuracy of the proposed assistance for discovering AR gestural interactions for their tasks. Based on these results, I aim to address issues related to interaction design and onboarding and design AR interactions that better follow fundamental HCI design principles (e.g., cognitive principles for design, Norman-Nielsen (NN) principles of interaction design [44, 45]) relevant to gestural interfaces.

4 EXPECTED CONTRIBUTION

Through my dissertation, I aim to contribute novel AR onboarding that will bridge the gaps between novice users' mental models and AR user interfaces, empowering users from diverse backgrounds and skill levels to comprehend and utilize AR applications more effectively. Overall, my dissertation aims to contribute:

- empirical insights into how users discover gestural input and interactions with AR applications and the importance of understanding interaction challenges to inform the design of AR interfaces;
- insights from experts that confirm discoverability issues with gestural input and identify key opportunities for AR researchers and designers to improve the onboarding experiences of novices and help them form an accurate mental model of the interaction;
- examine how software users employ new-generation LLM assistants to seek software help, both with and without prompt guidelines as well as integrated software context, and illustrate the challenges that software users experience when employing LLM optimized for software assistance;
- design and implementation of onboarding in AR environments that aids gesture discoverability for novice users. Empirical insights into the strengths and weaknesses of the proposed onboarding based on users' perceptions of usefulness, accuracy, and relevancy in assisting users in discovering correct gestural interactions for their tasks.

5 BIOGRAPHICAL SKETCH

Anjali Khurana is a Ph.D. student at Simon Fraser University (SFU), Burnaby, advised by Prof. Parmit K Chilana. She began her Ph.D. program in September 2021, has passed her Qualifying Examination and is working towards her proposal and expected to graduate in

2025. Her research interest lies in understanding users' mental models when employing novel interaction paradigms, such as gestural interactions with AR and prompt-based interactions with LLMs for their software tasks, and further inventing onboarding and assistance techniques for novice users of AR and LLMs. Her research has contributed a comprehensive understanding of how a diverse group of users begin working within unfamiliar AR environments and prompt-based interactions with LLMs. She will benefit from the feedback obtained from UbiComp/ISWC community through Doctoral Colloquium to design onboarding in AR environments that facilitates gesture discoverability for novice users. She has published her initial PhD research works as the lead author at top HCI venues including, ACM IUI 2024 [28], and Proc. ACM Interact. Mob. Wearable Ubiquitous Technol. (IMWUT 2023) [27].

ACKNOWLEDGMENTS

This research was supported by Natural Sciences and Engineering Research Council of Canada (NSERC) with award number: RGPIN-2020-06432.

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